The Impact of HLA-DRB1 Alleles on Anti-HLA Class I Antibody Formation

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Note: The results presented in this poster were reanalyzed in relation to the submitted abstract due to an MFI parameterization error in the original analysis, in which the cutoff was set to 11,000 instead of 1,000.

INTRODUCTION

IgG HLA antibody production depends on the presentation of HLA peptides by HLA class II molecules to CD4+ T-cells. The peptide binding to the HLA class II molecule is influenced by polymorphic sites located within the groove of the class II molecule.

In the present pilot study investigated the associations between class II HLA DRB1 alleles and antibody production against specific HLA class I targets. The strategy focused on targets that are highly prevalent in the general population.

OBJECTIVES

- To assess the association between individuals' HLA-DRB1 alleles and the formation of antibodies against HLA-A*01:01, A*02:01, and B*44:03.

METHODS

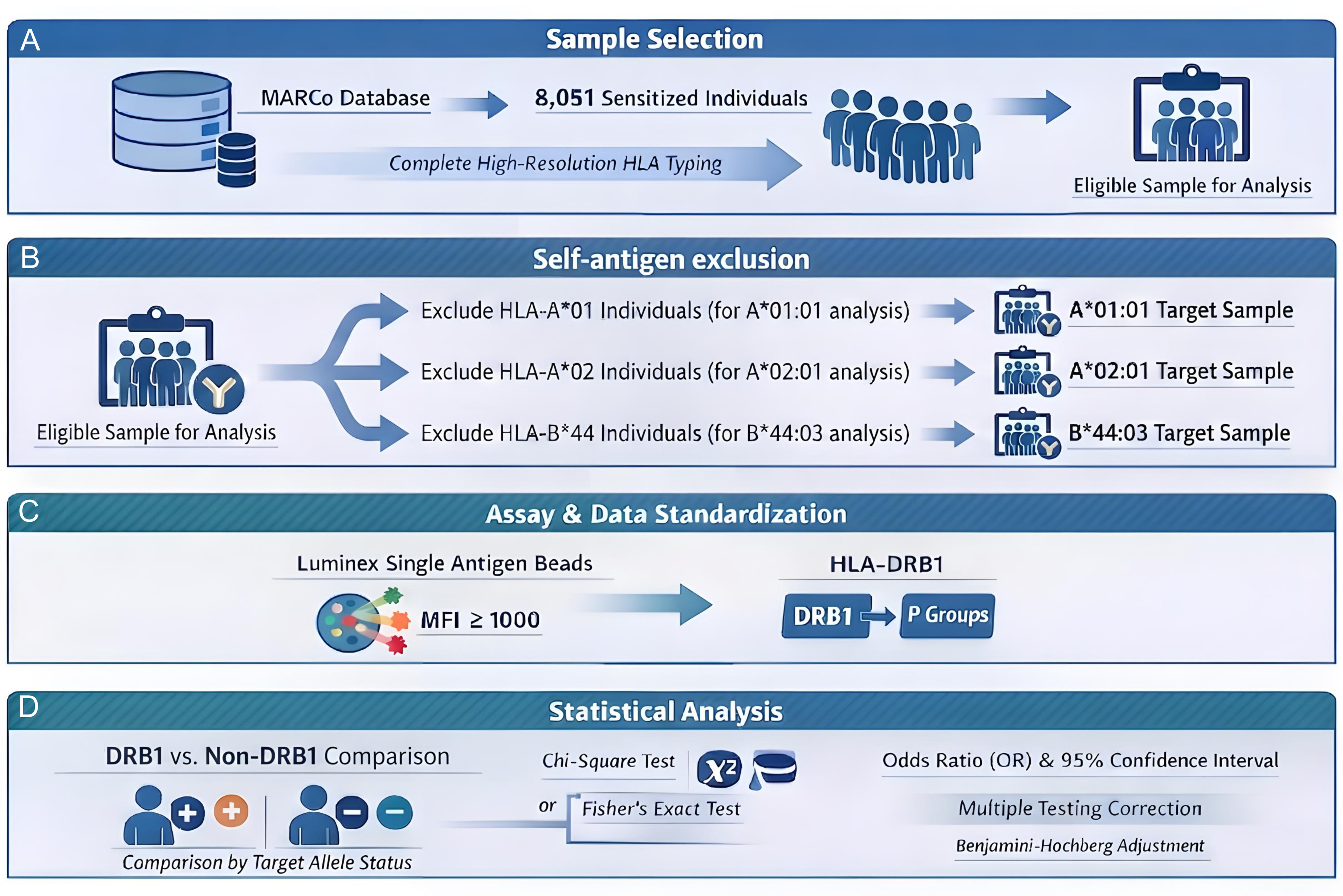

Figure 1. Study design and analytical workflow.

A. **Study population:** Luminex Single Antigen (LSA) results from MARCo (Molecular Antigen Reactivity Correlation; www.igen.org.br/marco) — an online platform developed by our laboratory that aggregates LSA assay results and maps HLA allele-pair reactivity correlations — were analyzed. The analysis comprised sera from 8,051 patients with prior sensitization events and high-resolution HLA typing.

B. **Self-antigen exclusion:** Individuals typed as HLA-A*01, HLA-A*02, and HLA-B*44 were excluded from the evaluations of antibodies against HLA-A*01:01, HLA-A*02:01, and HLA-B*44:03, respectively.

C. **Assay and standardization:** HLA antibody positivity was determined using LSA beads with an MFI threshold of ≥ 1,000, and HLA-DRB1 alleles were analyzed as P-groups.

D. **Statistical analysis:** The association between HLA-DRB1 alleles and antibody formation was analyzed using chi-square or Fisher's exact tests, with Benjamini-Hochberg-adjusted q-values used to determine significance.

RESULTS

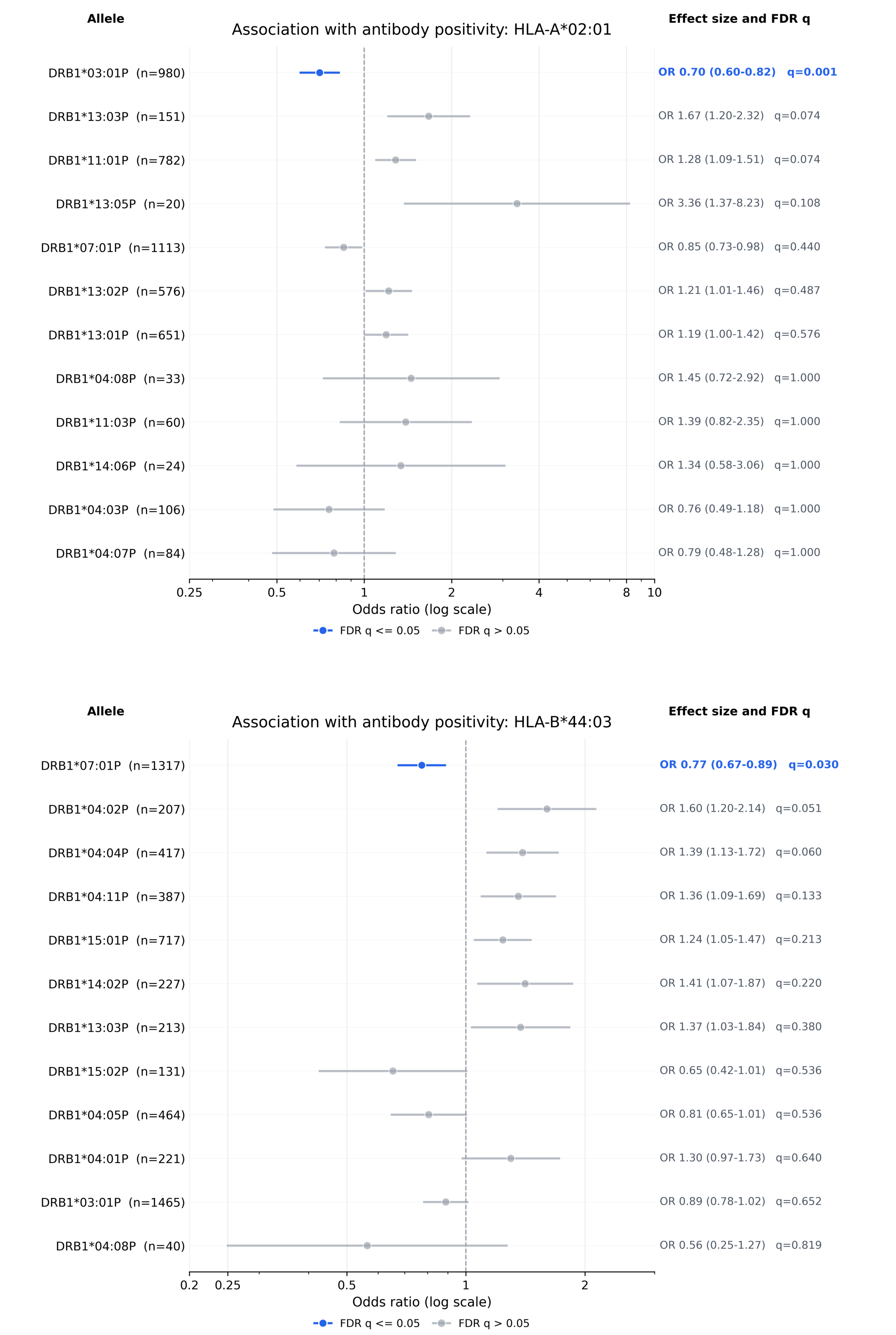

Figure 2. Forest plots of the association between HLA-DRB1 and antibody positivity against HLA-A*02:01 and HLA-B*44:03. Odds ratios (OR) and 95% confidence intervals are shown for the comparison of each HLA-DRB1 allele. The blue markers indicate associations that remained significant after Benjamini-Hochberg correction for multiple testing (FDR q ≤ 0.05), whereas gray markers indicate non-significant associations after correction. HLA-A*01:01 was not included in this figure because no HLA-DRB1 association remained significant after multiple-testing correction.

CONCLUSIONS

- Significant negative associations were observed between HLA-DRB1 alleles and antibody formation against HLA-A*02:01 and B*44:03.
- No significant association was observed between HLA-DRB1 alleles and antibody formation against HLA-A*01:01.
- Further studies are warranted to explore these findings.

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POSTER REFERENCE

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